NATIONAL UNIVERSITY OF SINGAPORE
DEPARTMENT OF ELECTRICAL & COMPUTER ENGINEERING

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SEMESTER 1

EE2213: Introduction to AI

TUTORIAL 5: L6

Question 1 (Lecture 6)

Which of the following functions are convex? Select all that apply.

(a) $f(x) = 5 - (x-2)^2, x \in \mathcal{R}$

(b) $f(x) = 7 - 3x,\ x \in \mathcal{R}$

(c) $f(x) = \log(x), x \in (0,\ +\infty)$

(d) $f(x) = e^x,\ x \in \mathcal{R}$

(e) $f(x) = |2x + 2|,\ x \in \mathcal{R}$

Ans: (b), (d), (e)

Question 2 (Lecture 6)

True or false: For a convex optimization problem with a differentiable objective function, gradient descent always converges to the global minimum if the learning rate is chosen appropriately.

(a) True

(b) False

Ans: (a)

Question 3 (Lecture 6)

Which of the following statements about gradient descent are correct? Select all that apply.

(a) Gradient descent moves in the direction of steepest descent, which is the negative gradient of the function.

(b) For convex functions with a properly chosen learning rate, gradient descent will converge to the global minimum.

(c) If the learning rate is too large, gradient descent may overshoot and fail to converge.

(d) Gradient descent is guaranteed to find the global minimum even for non-convex functions.

(e) Gradient descent only works for functions that are differentiable and strictly convex.

Ans: (a), (b), (c)

Question 4 (Lecture 6)

Suppose we are minimizing $f(x) = x^6$ with respect to x. We initialize $x = 1$, and perform gradient descent with learning rate $\eta = 0.01$.

(i) What is the updated value of x after the second iteration? Round your answer to 2 decimal places.

(ii) Explain why the updates in gradient descent become very small as x approaches 0. How does the shape of the function $f(x) = x^6$ influence the speed of convergence compared to a function like $f(x) = x^2$?

Ans:

(i) After the second iteration, the value of x is 0.90 (rounded to the 2 decimal places).

(ii) As x approaches 0, the derivative of the function, $f'(x) = 6x^5$, becomes very small. Since gradient descent updates are proportional to the gradient ($\mathbf{x}_{k+1} \leftarrow \mathbf{x}_k - \eta \nabla_x f(\mathbf{x}_k)$), the update steps also become very small.

The shape of $f(x) = x^6$ is very flat near $x = 0$, unlike $f(x) = x^2$, which has a steeper slope near the minimum. This flatness causes gradient descent to move slowly toward the minimum for $f(x) = x^6$, requiring many iterations to converge compared to a function like $f(x) = x^2$.

Question 5 (Lecture 6)

Suppose we have a function that depends on a vector of numbers $\mathbf{w}$:

$$C(\mathbf{w}) = \sum_{i=1}^{m} (\mathbf{w}^T \mathbf{x}_i - y_i)^2$$

where:

$\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_m$ are given vectors,

$y_1, y_2, \dots, y_m$ are given numbers (i.e., scalars).

We want to minimize $C(\mathbf{w})$ using gradient descent.

(i) Check if the function $C(\mathbf{w})$ is convex. Explain your reasoning.

(ii) Compute the gradient of $C(\mathbf{w})$ with respect to $\mathbf{w}$ using $\sum$ notation.

(iii) Write the gradient descent update rule for $\mathbf{w}$ with learning rate η.

(iv) If $\mathbf{x}_1 = \begin{bmatrix}1\\0\end{bmatrix}$, $\mathbf{x}_2 = \begin{bmatrix}0\\1\end{bmatrix}$, $\mathbf{x}_3 = \begin{bmatrix}1\\1\end{bmatrix}$, $\mathbf{y} = \begin{bmatrix}y_1\\y_2\\y_3\end{bmatrix} = \begin{bmatrix}2\\3\\5\end{bmatrix}$, $\eta = 0.01$, we initialize $\mathbf{w} = \begin{bmatrix}0\\0\end{bmatrix}$, calculate the updated value of $\mathbf{w}$ after the first iteration.

(v) (*Optional*): Following the setup in (iv), solve the convex optimization problem using a solver (e.g., CVXPY).

Ans:

(i) Each term $(\mathbf{w}^T\mathbf{x}_i - y_i)^2$ is a squared linear function of $\mathbf{w}$. Squaring a linear function always gives a convex function. The sum of convex functions is also convex. Therefore, $C(\mathbf{w})$ is convex.

(ii) Gradient of $C(\mathbf{w})$ with respect to $\mathbf{w}$:

$$\nabla_{\mathbf{w}} C(\mathbf{w}) = 2\sum_{i=1}^{m}(\mathbf{w}^T\mathbf{x}_i - y_i)\mathbf{x}_i$$

(iii) Gradient descent update rule:

$$\mathbf{w}_{k+1} = \mathbf{w}_k - \eta\nabla_{\mathbf{w}} C(\mathbf{w}) = \mathbf{w}_k - 2\eta\sum_{i=1}^{m}(\mathbf{w}^T\mathbf{x}_i - y_i)\mathbf{x}_i$$

(iv) After first iteration: $\mathbf{w}_{new} = \begin{bmatrix} 0.14 \\ 0.16 \end{bmatrix}$

You can solve it by hand, or refer to *TUT5-Q5-iv.py* for sample Python implementation.

(v) (Optional):

$$\mathbf{w}^* = \begin{bmatrix} 2 \\ 3 \end{bmatrix}, \qquad C(\mathbf{w}^*) = 0$$

where $\mathbf{w}^*$ is the optimal vector that minimizes the function $C(\mathbf{w})$.

Refer to *TUT5-Q5-v.py* for sample Python implementation.